B4.1 Adaptation to environment

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b4-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 341–362 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Habitats and adaptations

B4.1.1 Habitat

pp. 341–342

A **habitat** is the place where an organism, population, species or community lives. A description can include geographical and physical location, and the type of ecosystem.

B4.1.2 Adaptations to the abiotic environment

pp. 342–344

Plant	Challenges	Adaptations
Marram grass, sand dunes	Little water, salt spray, wind	Rolled leaves, sunken stomata, thick cuticle, deep roots and rhizomes
Mangrove trees, swamps	Anaerobic, salty, soft mud	Pneumatophores, stilt roots, salt glands

B4.1.3 Abiotic variables and distribution

pp. 344–345

Plants are affected by temperature, water, light, soil pH, salinity and mineral nutrients. Animals are affected by temperature, water and breeding sites, among others. A species' adaptations give it a range of tolerance.

B4.1.4 Range of tolerance

pp. 345–352

Each species thrives in an **optimum** range, is under stress towards the limits, and cannot survive beyond them. A **transect** lets you correlate the distribution of a species with an abiotic variable, using data you collect yourself. A correlation alone does not prove cause.

Reefs and biomes

B4.1.5 Conditions for coral reefs

p. 353

Conditions for coral reefs

B4.1.5

Factor	Condition	Why
Depth	Mostly under 25 m	Light reaches algae
pH	Above 7.8	Calcium carbonate can form
Salinity	About 35 parts per thousand	Avoids osmotic stress
Clarity	Clear water	Light passes through
Temperature	About 23–29 °C	Suits corals and algae

Threats
Warming causes bleaching; ocean acidification lowers pH.

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Reefs need shallow, clear, warm water with suitable pH and salinity. Most reef-building corals live less than 25 m deep, at about 23–29 °C. Corals have a mutualistic relationship with algae, the zooxanthellae, which need light.

B4.1.6 Climate determines land biomes

pp. 354–355

For any given temperature and rainfall pattern, one type of natural ecosystem is likely to develop. A graph of these two variables shows where each biome is found.

B4.1.7 Similar communities in each biome

pp. 356–359

Similar abiotic conditions produce similar selection pressures, and so **convergent evolution** gives similar communities. Know the climates of tropical forest, temperate forest, taiga, grassland, tundra and hot desert.

B4.1.8 Hot deserts and tropical rainforests

pp. 359–361

Species	Biome	Adaptations
Saguaro cactus	Hot desert	Water-storing stem, spines, shallow wide roots
Fennec fox	Hot desert	Large ears lose heat, active at night
Spider monkey	Rainforest	Long limbs and a gripping tail
Poison dart frog	Rainforest	Toxins and warning colours

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